

TINGJUN LI

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RESEARCH PROFILE

Doctoral researcher specialising in autonomous-driving decision-making, socially compatible driver behaviour modelling, multi-agent inverse reinforcement learning, trust-aware interaction, and vehicle energy management. Experienced in end-to-end research from human-driving data analysis and algorithm development to software-in-the-loop and human-in-the-loop validation. Author of two papers in *IEEE Transactions on Intelligent Transportation Systems* and recipient of the CVCi 2025 Kong Hui Paper Award.

EDUCATION

PhD in Vehicle Engineering, College of Automotive Engineering, Jilin University, China (Sep 2019–Mar 2027)
Expected

BEng in Vehicle Engineering, College of Automotive Engineering, Jilin University, China (Sep 2015–Jul 2019)

AWARDS AND HONOURS

- Kong Hui Paper Award, 9th CAA International Conference on Vehicular Control and Intelligence (CVCi), 2025.
- Best Speaker Award in the Cambridge Academic Programme For School of Mechanical and Aerospace Engineering Jilin University, 2021.
- Mathematical Contest in Modeling, Meritorious Winner (top 10%), 2018; First Prize, Jilin Provincial Mathematical Modeling Competition, 2017.
- Graduate Academic Scholarships; National Encouragement Scholarship; Outstanding Student Award.

SELECTED PUBLICATIONS

1. **T. Li**, N. Xu, S. Feng, H. Askari, B. H. G. Barbosa, and K. Guo, “Responsibility-Based Socially Compatible Driving Behavior Modeling Verified by Hierarchical Multi-Agent Inverse Reinforcement Learning,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 26, no. 9, pp. 14353–14370, 2025. doi:10.1109/TITS.2025.3577660.
2. **T. Li**, N. Xu, S. Feng, H. Askari, B. H. G. Barbosa, and K. Guo, “Pride and Prejudice: Toward an Information-Theoretic Framework for Mutually Communicative Driver Behavior Modeling,” *IEEE Transactions on Intelligent Transportation Systems*, 2026. doi:10.1109/TITS.2026.3703952.
3. **T. Li**, N. Xu, and K. Guo, “Modelling Trust with Interaction Uncertainty Verified by Mutual Subjective Evaluation,” in *2025 9th CAA International Conference on Vehicular Control and Intelligence (CVCi)*, Qindao, China, pp. 1–6, 2025. **Kong Hui Paper Award.**
4. **T. Li**, N. Xu, and K. Guo, “Deep Reinforcement Learning Based Truck Eco-Driving in Mixed Traffic Utilizing Terrain Information,” in *Proceedings of China SAE Congress 2021*, pp. 183–191.

RESEARCH EXPERIENCE

Risk Perception, Quantitative Assessment and Safe Decision-Making for Highly Automated Vehicles in Complex Traffic, National Key R&D project (Feb 2023–Mar 2026)

Subtask Lead

- Developed socially compatible game-theoretic interaction models and learned driving styles from NGSIM lane-change data; designed a multi-agent inverse reinforcement learning framework for model calibration.
- Conducted statistical analysis of social compatibility in NGSIM interactions; the work resulted in two first-quartile *IEEE T-ITS* papers.
- Designed an interaction-trust measurement method and validated it through human-in-the-loop simulation; the resulting paper received the CVCi 2025 Kong Hui Paper Award.
- Contributed 6 of 23 technical definitions to group standard T/ITS 0302-2025, *Terminology and Definitions for Risk Perception and Assessment Technology of Automated Vehicles*.
- Developed reinforcement-learning methods for trustworthy interactive decision-making.

Dongfeng RoboTaxi L4 Autonomous-Driving Development, Industry-sponsored project (Dec 2020–Jul 2022)

Decision Module Lead

- Developed decision algorithms using decision trees and the COSIM lane-change model for road-segment and intersection planning; proposed a free–protected–forced three-stage pre-intersection model.

- Delivered a scenario-based workflow spanning rule design, MATLAB/Simulink development, ROS/C++ reimplementation, and SCANeR software-in-the-loop validation.

Coordinated Active-Safety Control and Application Validation under Extreme Conditions, Major research project (Sep 2020–Dec 2021)

Researcher

- Investigated tyre-pressure regulation, tyre slip, friction, and optimisation-based energy-saving control.
- Supported drive-by-wire chassis testing and processed GPS/IMU module data.

Intelligent Vehicle–Traffic-Signal Coordination in Mixed Traffic at Ramps, Research project (Sep 2019–Dec 2021)

Principal Investigator

- Reviewed cooperative control methods for mixed traffic comprising autonomous and human-driven vehicles.
- Developed cooperative eco-driving control with deep reinforcement learning, improved the commercial-vehicle energy model, and published the work at China SAE Congress 2021.

Intelligent Energy Management for Electric Vehicles Based on Road-Condition Prediction, Joint-fund project (Sep 2019)

Researcher

- Built a vehicle model combining a seven-degree-of-freedom model with the UniTire tyre model and designed a heuristic slip-energy optimiser for tyre-pressure control.
- Supported power-supply, CAN-bus, and structural design for a four-wheel-drive platform, including GPS/IMU calibration and functional validation.

TEACHING AND SERVICE

College of Automotive Engineering, Jilin University (Sep 2021–Jun 2022)

Student Tutor

- Delivering a practical and tutorial lecture for courses in Patent Search and Analysis for groups of up to 100 graduate students
- Training undergraduate students and junior lab members and supervising thesis projects to ensure competence.

Student Union, College of Automotive Engineering, Jilin University (Sep 2017–Jun 2018)

Deputy Director

- Organised and hosted academic guidance lectures (approximately 300 attendees) and experience-sharing events (approximately 200 attendees); trained students in event scripts, news releases, and publicity writing.

TECHNICAL AND LANGUAGE SKILLS

Programming	Python; working knowledge of R; ROS/C++
Simulation	MATLAB/Simulink, CarSim, SCANeR, LabVIEW, CAD, CATIA
Research methods	Reinforcement learning, inverse reinforcement learning, multi-agent modelling, driver behaviour modelling, human-in-the-loop and software-in-the-loop validation
Languages	Chinese (native); English (TOEFL 100; CET-6 565; CET-4 574)
Training	Coursera courses in Machine Learning and Behaviour Planning; National Computer Rank Examination Level 2